DS3231 + AT24C32 RTC Breakout

High-Precision Real-Time Clock & 32Kb I²C EEPROM Module (Model ZS-042)

I²C BUS | TCXO ±2PPM | 3.3V – 5.5V

1. MODULE OVERVIEW & FEATURES

The **ZS-042** breakout board combines the Maxim/Analog Devices **DS3231** high-precision Real-Time Clock with an Atmel/Microchip **AT24C32** 32-Kbit EEPROM memory chip. It features an integrated Temperature-Compensated Crystal Oscillator (TCXO) with an internal 32.768 kHz resonator, maintaining precision across temperatures.

- **Clock Accuracy:** ±2 ppm (0°C to +40°C), ±3.5 ppm (-40°C to +85°C)
- **Operating Voltage:** 2.3V to 5.5V DC (3.3V and 5.0V compatible)
- **Active Current:** ~200 µA; **Timekeeping Standby:** ≤1.5 µA
- **Memory:** 32 Kbit (4,096 x 8-bit) non-volatile I²C EEPROM
- **Calendar Features:** Seconds, minutes, hours, day, date, month, year with leap-year compensation valid up to year 2100
- **Alarms & Waveform:** 2 programmable alarms, 32.768 kHz & configurable square-wave interrupt outputs

2. PINOUT & HEADER SPECIFICATIONS

Pin	Type	Function & Description
`32K`	Output	32.768 kHz TCXO output (open-drain; needs external pull-up)
`SQW`	Output	Active-low alarm interrupt or programmable square-wave
`SCL`	Input	I²C Clock line (onboard 4.7 kΩ pull-up to VCC)
`SDA`	I/O	I²C Data line (onboard 4.7 kΩ pull-up to VCC)
`VCC`	Power	Main power supply: **3.3V to 5.5V DC**
`GND`	Power	System ground reference (0V)

I²C Slave Addresses:
- **DS3231 RTC:** `0x68` (Fixed)
- **AT24C32 EEPROM:** `0x57` (Default; address pads A0, A1, A2 selectable from 0x50 to 0x57)

3. PHYSICAL DIMENSIONS & MECHANICAL LAYOUT

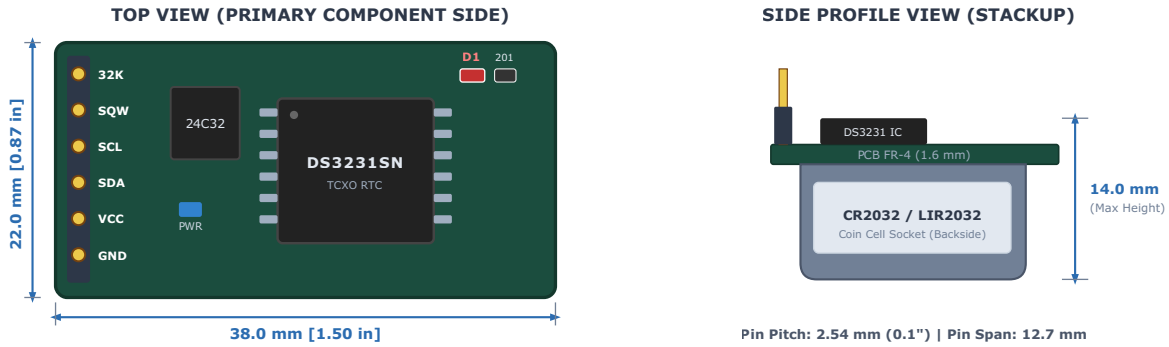


Figure 1: ZS-042 Mechanical Dimensioning, Component Placements, and Side Profile Stackup.

Key Technical References & Datasheet Links:
- **Maxim / Analog Devices DS3231 RTC IC:** https://www.analog.com/media/en/technical-documentation/data-sheets/DS3231.pdf
- **Microchip AT24C32D 32Kb I²C EEPROM:** https://ww1.microchip.com/downloads/a5/documents/MEE/DataSheet/AT24C32D-32-Kbit-I2C-Serial-EEPROM-20006047B.pdf
- **3D STEP CAD Model:** Saved in project repository at `models/ds3231_rtc.step`

DS3231 Module Circuit & Battery Modification

CR2032 vs LIR2032 Safe Operation & Schematic Analysis

HARDWARE SAFETY GUIDE

CRITICAL BATTERY SAFETY NOTICE: DIODE MODIFICATION REQUIRED FOR CR2032

The ZS-042 breakout board comes factory-wired with an active trickle-charging circuit designed *only* for **rechargeable lithium-ion LIR2032** cells (3.6V nominal, 4.2V max). If you install a standard, non-rechargeable **CR2032 Lithium Manganese Dioxide** coin cell (3.0V primary battery) and power the module with 5V, continuous charging current will be forced through diode **D1** and resistor **R1 (201)** directly into the non-rechargeable cell. Over time, this causes electrolyte boiling, gas generation, battery swelling, chemical leakage, and potential cell rupture.

4. BATTERY TYPES COMPARISON

Parameter	CR2032 (Standard)	LIR2032 (Rechargeable)
Chemistry	Li-MnO_2 (Primary / Disposable)	Li-Ion (Secondary / Rechargeable)
Nominal Voltage	3.0 V (Fresh: 3.2 V)	3.6 V (Peak Charge: 4.2 V)
Capacity	~210 – 240 mAh	~35 – 70 mAh
Shelf Life	8 – 10 years	Self-discharges in ~2–4 months
Recharging	**STRICTLY FORBIDDEN**	Required
Recommended for RTC?	**YES (with diode removed)**	No (poor longevity in RTCs)

5. HOW TO MODIFY BOARD FOR CR2032

1. Locate diode **D1** (labeled with red line/triangle, adjacent to resistor marked **201**) near the positive battery terminal pad.
2. **Method A (Recommended):** Touch a soldering iron tip to both ends of **D1** and flick it off the PCB pads.
3. **Method B:** Snip or crush diode **D1** in half using micro flush cutters to break the circuit.
4. *Alternative:* Alternatively desolder or remove resistor **201** (200 Ω).

6. DIODE LOCATOR DIAGRAM

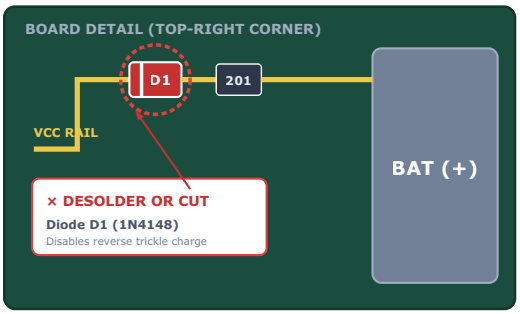


Figure 2: Component Locator for Diode D1 and Resistor 201 (200 Ω).

7. COMPLETE MODULE SCHEMATIC (ZS-042 CIRCUIT DIAGRAM)

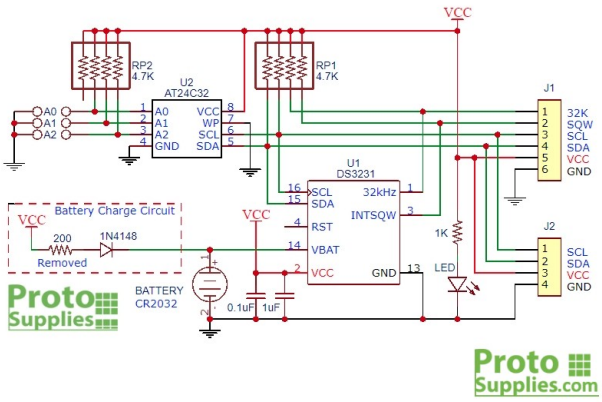


Figure 3: Full ZS-042 schematic depicting DS3231 RTC, AT24C32 EEPROM, I²C pull-up network, and the VCC-to-VBAT charging branch.